

Branch-Normalized Partial Returns in Single-Defect Circular Automata

WuJun Chen
Independent Researcher
2026

This paper (Paper XXXI of the RIME program) studies intrinsic raw spectator-safe return dynamics. It imports the missing-image section and branch freedom proved in Paper XXX, but not any typed transfer, projectability, recursive-return, or credit-settlement conclusion.

Abstract

Problem. Paper XXX shows that every labelled binary-kernel rank- $(n - 1)$ defect has a universal return group on its missing-image section, and that after fixing the oriented kernel and missing image the remaining branch is an arbitrary permutation. The group alone does not retain the guards of the other partial returns. The resulting Safe-Hit problem therefore still depends on a partial dynamical system.

Approach. We conjugate the entire labelled return system, not only its two globally enabled generators. For kernel separation Δ and branch permutation a , every normalized return has point map

$$\phi_r = \varepsilon_\Delta \circ p^r \circ a,$$

with an exact collision guard and an exact normalized exit target. We first eliminate the exponent in the cyclic branch $a \in \langle \psi_\Delta \rangle$. We then decompose the punctured rotation ψ_Δ into arithmetic lanes and use a guarded hole-slide to classify target reachability. Finally, we record the part of this classification that survives for branch permutations normalizing $\langle \psi_\Delta \rangle$.

Results. The branch conjugacy transports guards, targets, and labelled paths exactly. In the cyclic branch, Safe-Hit truth is independent of the branch exponent. Put $g = \gcd(n, \Delta)$ and read each ψ_Δ -lane after deleting holes. A marked pair reaches the section exit target if and only if its two tokens lie in one lane and are adjacent in that hole-deleted token order. Thus, for every integer s ,

$$\text{SafeHit}_{n,\Delta,\psi_\Delta^s}(x) \iff \text{LaneAdj}_\Delta(x) = 1.$$

For a normalizer branch, the quotient dynamics is an exact relation-valued skew system: each edge is the identity-branch relation preceded by a deterministic orbit permutation. Normalizer elements admit an affine lane form with orientation-compatible multiplier $u \equiv \pm 1$ (lane-order preserving or reversing), for which the same lane-adjacency criterion is complete. For every normalizer branch we obtain the sandwich

$$\text{LaneAdj}_\Delta(x) \implies \text{SafeHit}_{n,\Delta,a}(x) \implies \text{SameLane}_\Delta(x).$$

Boundary. The paper does not classify multipliers $u \neq \pm 1$, the broader cycle-partition-preserving regime, or terminal survivor placement. It proves raw marked-pair reachability only. It does not establish transfer membership, authorization, projectability, recursive return, credit settlement, or a reset bound.

Keywords. synchronizing automata; circular automata; rank defect; partial dynamics; Safe-Hit; punctured rotation; lane adjacency; normalizer

Notation Table

Symbol	Meaning
$Q = \mathbb{Z}/n\mathbb{Z}$	labelled circular carrier, $n \geq 6$
$p(q) = q + 1$	fixed full cycle
$d : Q \rightarrow Q$	rank- $(n - 1)$ map with one binary kernel
$K_d = \{0, \Delta\}$	oriented kernel after cyclic relabelling
m	unique missing image of d
$D = Q \setminus \{m\}$	missing-image section carrier before normalization
$E = Q \setminus \{0\}$	normalized branch carrier
$b = d _E : E \rightarrow D$	inverse-branch bijection
t_0	exponent satisfying $p^{t_0}(m) = 0$
$a = p^{t_0}b$	branch permutation in $\text{Sym}(E)$
ε_Δ	marked collapse $0 \mapsto \Delta$, fixing E
$\widehat{\Sigma} = \text{Conf}_{2,3}(E)$	marked pair plus three spectators
U_x	five-point occupied support of x
$\phi_r = \varepsilon_\Delta p^r a$	normalized partial return point map
$\widehat{\mathcal{T}}_{\Delta,a}$	normalized section exit target
$\psi_\Delta = \varepsilon_\Delta p^\Delta _E$	punctured rotation
$H_\Delta = \langle \psi_\Delta \rangle$	universal relative-return subgroup
$C_0, C_1, \dots, C_{g-1}$	punctured and ordinary ψ_Δ -lanes
LaneAdj_Δ	same-lane adjacency after deleting holes
SameLane_Δ	both marked tokens lie in one lane

1 Introduction

A rank- $(n - 1)$ defect has one binary collision fiber, but its return dynamics need not be local. Away from the kernel, the labelled defect may carry the occupied coordinates by an arbitrary bijection. Paper XXIX classifies the complete raw rank-five first-exit frontier for the canonical defect by a cyclic token order and a constructive hole slide [4]. Paper XXX then shows that every labelled defect has two globally enabled section returns, an intrinsic punctured rotation, and a free branch permutation [5].

The latter result changes the natural question. The universal return group describes globally available symmetry, but it does not determine which of the remaining return blocks are enabled on a given five-point support. The next object is therefore the complete partial return system. After orienting the kernel and conjugating by one inverse branch, this system has the exact normal form

$$\boxed{\phi_r = \varepsilon_\Delta p^r a.} \quad (1.1)$$

Equation (1.1) transports the guard, the section target, and every labelled path. It reduces the raw problem for an arbitrary labelled defect to the parameters (n, Δ, a) without erasing the collision semantics.

The paper's main theorem concerns the structured cyclic branch $a \in \langle \psi_\Delta \rangle$. The cycles of the punctured rotation form arithmetic lanes. The guarded maps preserve the hole-deleted token order in each lane, while a specific partial return moves a hole past the marked pair and strictly decreases a finite gap count. This proves the path-free classification

$$\boxed{a \in \langle \psi_\Delta \rangle \implies \text{SafeHit}_{n,\Delta,a}(x) \iff \text{LaneAdj}_\Delta(x) = 1.} \quad (1.2)$$

We then ask how far (1.2) survives beyond the cyclic branch. If a normalizes $H_\Delta = \langle \psi_\Delta \rangle$, it acts by a deterministic twist on the H_Δ -orbit space. The other returns remain relations on that quotient. An affine lane normal form isolates a multiplier u , a lane permutation, and phase data. The classification survives exactly as proved here for the orientation-compatible multipliers $u \equiv \pm 1$. For the full normalizer, lane adjacency is sufficient and same-lane incidence

is necessary. The unresolved region is therefore confined to same-lane, nonadjacent marked pairs under multipliers $u \neq \pm 1$.

The theorem spine is short.

1. Section 2 proves the exact branch-normalized partial system.
2. Section 3 eliminates cyclic-branch exponents at the level of Safe-Hit truth.
3. Section 4 proves the multi-lane invariant, the terminating hole slide, and the complete cyclic-branch classification.
4. Section 5 gives the normalizer skew quotient and affine lane form.
5. Section 6 closes the multipliers $u \equiv \pm 1$ and localizes the open full-normalizer frontier.

The work remains at the intrinsic raw level. In particular, a raw Safe-Hit path is not silently promoted to a typed transfer, an authorized entry, or a projection witness.

1.1 Related Work and Novelty Boundary

The Černý conjecture asks for a quadratic reset bound for synchronizing automata [3, 12, 13]. Circular automata admit stronger reset results under hypotheses different from those used here [7]. Almost-group automata, built from permutation letters and a letter with one nontrivial kernel class, form a nearby structural setting [1, 2]. This paper proves no reset bound and does not classify general almost-group automata.

Monotonic and cyclic-order methods study automata preserving a global order [8]. Subset and careful synchronization, as well as synchronization under dynamic constraints, study synchronizing subsets or words subject to partiality, order, or visitation constraints [11, 14]. Complete reachability studies which unlabelled subsets arise as images [6, 9, 10]. The present observable is different: one guarded path must preserve injectivity of the marked pair and all three spectators at every proper return step, and that same path must hit a source-marked target. The cited results do not supply the lane-adjacency or normalizer-skew classifications proved here. The novelty is the exact branch normalization of this partial system and the resulting multi-lane, source-marked Safe-Hit classification.

Within the RIME line, Paper XXIX supplies the canonical single-lane model and Paper XXX supplies the missing-image section, universal return group, punctured rotation, and branch freedom [4, 5]. The present paper does not repeat their frontier or group classifications. It uses their interfaces to classify a new structured family of partial return systems.

2 Branch Normalization and the (n, Δ, a) System

Let $n \geq 6$, let

$$Q = \mathbb{Z}/n\mathbb{Z}, \quad p(q) = q + 1 \pmod{n}, \quad (2.1)$$

and let $d : Q \rightarrow Q$ have rank $n - 1$, one binary kernel, and missing image m . Orient the kernel and relabel cyclically so that

$$K_d = \{0, \Delta\}, \quad 1 \leq \Delta \leq n - 1. \quad (2.2)$$

Put

$$D = Q \setminus \{m\}, \quad E = Q \setminus \{0\}, \quad (2.3)$$

and let t_0 be the unique exponent with $p^{t_0}(m) = 0$. The restriction

$$b = d|_E : E \xrightarrow{\sim} D \quad (2.4)$$

is a bijection. Define

$$a = p^{t_0} \circ b \in \text{Sym}(E) \quad (2.5)$$

and the marked collapse

$$\varepsilon_\Delta(0) = \Delta, \quad \varepsilon_\Delta(q) = q \quad (q \neq 0). \quad (2.6)$$

Its only non-singleton fiber is $\{0, \Delta\}$, and

$$\boxed{b^{-1}d = \varepsilon_\Delta.} \quad (2.7)$$

We use the automata convention that the written block $p^t d$ acts first by p^t and then by d , so its point map is $d \circ p^t$.

The normalized configuration space is

$$\widehat{\Sigma} = \text{Conf}_{2,3}(E) = \{(A, R) : |A| = 2, |R| = 3, A \cap R = \emptyset\}. \quad (2.8)$$

For $x = (A, R)$ write $U_x = A \cup R$. The branch bijection acts on states by

$$\mathbf{b}(A, R) = (bA, bR), \quad \mathbf{b} : \widehat{\Sigma} \xrightarrow{\sim} \Sigma_m, \quad (2.9)$$

where Σ_m is the missing-image section of Paper XXX. If $\mathcal{R}_t = d \circ p^t$ is the corresponding partial section return, re-index its branch conjugate by

$$\widetilde{\mathcal{R}}_r = \mathbf{b}^{-1} \mathcal{R}_{t_0+r} \mathbf{b}. \quad (2.10)$$

This notation means conjugation of a partial state map, including its domain.

Theorem 2.1 (branch-normalized full return dynamics; N1).

Define

$$\phi_r = \varepsilon_\Delta \circ p^r \circ a : E \rightarrow E. \quad (2.11)$$

Then for every $r \in \mathbb{Z}/n\mathbb{Z}$:

1. *the conjugated return is*

$$\boxed{\widetilde{\mathcal{R}}_r = (\phi_r)_\#;} \quad (2.12)$$

2. *its exact guard is*

$$\boxed{x \in \text{Dom}(\widetilde{\mathcal{R}}_r) \iff \{0, \Delta\} \not\subseteq p^r a(U_x);} \quad (2.13)$$

3. *under this guard,*

$$\widetilde{\mathcal{R}}_r(A, R) = (\phi_r(A), \phi_r(R)); \quad (2.14)$$

4. *the normalized inverse image of the section exit set is*

$$\boxed{\widehat{\mathcal{T}}_{\Delta,a} = \{(A, R) \in \widehat{\Sigma} : \exists u, p^u a(A) = \{0, \Delta\}\};} \quad (2.15)$$

5. *after the label translation $r \mapsto t_0 + r$, $\mathbf{b}$ is an isomorphism of partial labelled transition systems. In particular,*

$$x \xrightarrow{r_1, \dots, r_j}_{\text{norm}} y \iff \mathbf{b}(x) \xrightarrow{t_0+r_1, \dots, t_0+r_j}_{\text{section}} \mathbf{b}(y). \quad (2.16)$$

Proof. The point map beneath (2.10) is

$$\begin{aligned} b^{-1}d p^{t_0+r}b &= \varepsilon_\Delta p^r(p^{t_0}b) \\ &= \varepsilon_\Delta p^r a. \end{aligned} \quad (2.17)$$

Only ε_Δ can destroy injectivity. It does so on U_x exactly when the preceding permutation $p^r a$ places two occupied points at 0 and Δ . This proves (2.12)–(2.14).

The original section target consists of states whose marked pair can be rotated onto K_d . Thus $\mathbf{b}(A, R)$ is a target exactly when some u satisfies $p^u b(A) = \{0, \Delta\}$. Replacing u by $u - t_0$ and using $a = p^{t_0}b$ gives (2.15), reversibly. Equation (2.17) identifies each single labelled edge, and induction on path length gives (2.16).

The normalized Safe-Hit predicate is therefore

$$\boxed{\text{SafeHit}_{n,\Delta,a}(x) \iff \exists r_1, \dots, r_j : x \xrightarrow{r_1, \dots, r_j}_{\text{norm}} \widehat{\mathcal{T}}_{\Delta,a}.} \quad (2.18)$$

All guards and the target are now functions of (n, Δ, a) . Paper XXX proves that every $a \in \text{Sym}(E)$ occurs in the fixed oriented kernel and missing-image fiber. Hence Theorem 2.1 classifies the input data for the complete partial dynamics. It does not yet classify reachability.

Two globally enabled returns are visible immediately. Put

$$\psi_\Delta = \varepsilon_\Delta p^\Delta|_E. \quad (2.19)$$

Then

$$\widetilde{\mathcal{R}}_0 = a_\#, \quad \widetilde{\mathcal{R}}_\Delta = (\psi_\Delta a)_\#. \quad (2.20)$$

The group skeleton $\langle a, \psi_\Delta \rangle$ is therefore the globally enabled part of (2.11); the remaining labels retain information that this group forgets.

3 Cyclic-Branch Exponent Elimination

Let

$$H_\Delta = \langle \psi_\Delta \rangle, \quad \pi_H : \widehat{\Sigma} \rightarrow \widehat{\Sigma}/H_\Delta. \quad (3.1)$$

For $s \in \mathbb{Z}$, set $a_s = \psi_\Delta^s$ and write

$$F_r^{(s)} = (\varepsilon_\Delta p^r \psi_\Delta^s)_\#, \quad \widehat{\mathcal{T}}_s = \widehat{\mathcal{T}}_{\Delta, \psi_\Delta^s}. \quad (3.2)$$

For H_Δ -orbits O, O' , define the existential quotient edge

$$O \xrightarrow{r}_s O' \iff \exists x \in O \cap \text{Dom}(F_r^{(s)}), \quad F_r^{(s)}(x) \in O'. \quad (3.3)$$

The quotient is relation-valued: no representative-independent partial map is assumed.

Theorem 3.1 (cyclic-branch exponent elimination; N2).

For every integer s :

$$\boxed{O \xrightarrow{r}_s O' \iff O \xrightarrow{r}_0 O'} \quad (3.4)$$

for all O, O', r ,

$$\boxed{\pi_H(\widehat{\mathcal{T}}_s) = \pi_H(\widehat{\mathcal{T}}_0),} \quad (3.5)$$

and, at every fixed starting state,

$$\boxed{\text{SafeHit}_{n,\Delta,\psi_\Delta^s}(x) \iff \text{SafeHit}_{n,\Delta,\text{id}}(x)}. \quad (3.6)$$

Proof. The globally enabled normalized returns are

$$F_0^{(s)} = (\psi_\Delta^s)_\#, \quad F_\Delta^{(s)} = (\psi_\Delta^{s+1})_\#,$$

and

$$\langle \psi_\Delta^s, \psi_\Delta^{s+1} \rangle = H_\Delta. \quad (3.7)$$

Their inverses are positive powers, so every H_Δ -orbit is strongly connected by actual return blocks in the s -system.

For an edge witness $x \in O$, put $y = (\psi_\Delta^s)_\#x$. Then $y \in O$ and

$$F_r^{(s)}(x) = F_r^{(0)}(y). \quad (3.8)$$

The two sides have the same guard because they present the same five-point set to $\varepsilon_\Delta p^r$. The substitution $x \mapsto y$ is a bijection of O , proving (3.4). Similarly,

$$x \in \widehat{\mathcal{T}}_s \iff (\psi_\Delta^s)_\#x \in \widehat{\mathcal{T}}_0,$$

which proves (3.5). Exact orbit lifting, using the internal controllability from (3.7), identifies each Safe-Hit predicate with reachability in its relation-valued orbit system. Equations (3.4)–(3.5) make the two quotient problems identical, proving (3.6).

The theorem compares truth predicates, not witness words. It does not assert equality of labelled path languages, shortest words, or deterministic quotient transitions.

4 Multi-Lane Invariant and Hole-Slide

It remains to solve the base system $a = \text{id}$. Put

$$g = \gcd(n, \Delta), \quad \ell = \frac{n}{g}, \quad L_c = \{q \in Q : q \equiv c \pmod{g}\}. \quad (4.1)$$

The cycles of ψ_Δ on E are the lanes

$$C_0 = L_0 \setminus \{0\}, \quad C_c = L_c \quad (c \neq 0). \quad (4.2)$$

The punctured lane C_0 has length $\ell - 1$; every ordinary lane has length ℓ . Each is read in its ψ_Δ -cyclic order. On C_0 this order contains the shortcut $-\Delta \mapsto \Delta$ that closes the deleted point.

For $x = (A, R)$, color the points of A by **P**, the points of R by **S**, and all holes in $E \setminus U_x$ by **H**. Read each lane cyclically and delete the holes. Define

$$\text{LaneAdj}_\Delta(x) = 1 \quad (4.3)$$

if and only if the two **P** tokens lie in one lane and are adjacent in that lane's hole-deleted cyclic word. Equivalently, one oriented lane arc between the marked tokens has no occupied point in its interior.

Lemma 4.1 (lane-order preservation; N3.1).

For every enabled base return $\phi_r = \varepsilon_\Delta p^r$, all lane indices undergo the same translation $c \mapsto c + r \pmod{g}$, and the hole-deleted cyclic **P/S** order in each occupied lane is preserved. Hence

$$\boxed{\text{LaneAdj}_\Delta((\phi_r)_\#x) = \text{LaneAdj}_\Delta(x)}. \quad (4.4)$$

Proof. Since $g \mid \Delta$, $\varepsilon_\Delta(q) \equiv q \pmod{g}$, and therefore $\phi_r(q) \equiv q + r \pmod{g}$. On the full Δ -cycles in Q , translation preserves cyclic order. Insert 0 as a hole when comparing a punctured lane with a full cycle, translate, and delete it again. The marked collapse closes the resulting gap. Its guard excludes simultaneous occupancy of 0 and Δ , so the insertion, deletion, and shortcut cannot identify or cross two occupied tokens. Deleting all holes leaves the same cyclic token word.

Lemma 4.2 (lane hole-slide; N3.2).

If $\text{LaneAdj}_\Delta(x) = 1$, then x reaches $\widehat{\mathcal{T}}_{\Delta, \text{id}}$ by guarded normalized returns.

Proof. Distinguish the marked tokens temporarily as P_1, P_2 . Choose an oriented lane arc from P_1 to P_2 with no occupied point in its interior, and let h be the number of holes in that arc.

If the marked lane is ordinary and $P_1 = z$, then either $h = 0$ and the marked pair is already a rotation of $\{0, \Delta\}$, or $z + \Delta$ is a hole. In the latter case, the label $r = -z$ presents the collision pair as $\{P, H\}$, so the return is enabled. It moves the marked lane into C_0 , removes the first hole from the chosen arc, and creates the compensating hole in a different lane. Thus h decreases by one.

Now suppose the marked pair lies in C_0 . A power of the globally enabled return ψ_Δ places P_1 at Δ without changing h . If $h > 0$, then 2Δ is a hole. The fixed label $r = -\Delta$ is enabled, fixes P_1 at Δ , moves the later points of the selected arc one Δ -step toward it, and creates the new hole at $-\Delta$, outside the selected forward arc. Hence

$$\boxed{h \mapsto h - 1}. \quad (4.5)$$

Iteration reaches $h = 0$. The marked pair is then a physical Δ -neighbor pair unless it occupies the punctured shortcut $\{-\Delta, \Delta\}$. In that case one ψ_Δ step sends it to $\{\Delta, 2\Delta\}$. The resulting pair is a rotation of $\{0, \Delta\}$, so the state already belongs to the declared target.

If $\ell = 2$, the punctured lane has one point and cannot contain both marked tokens. If $\ell = 3$, its two points are already adjacent. Thus the decreasing step is invoked only when the stated next coordinate exists.

5 Complete Cyclic-Branch Classification

Theorem 5.1 (multi-lane Safe-Hit classification; N3).

For every $n \geq 6$, $1 \leq \Delta \leq n - 1$, and $x \in \widehat{\Sigma}$,

$$\boxed{\text{SafeHit}_{n, \Delta, \text{id}}(x) \iff \text{LaneAdj}_\Delta(x) = 1}. \quad (4.6)$$

Proof. Lemma 4.1 makes lane adjacency invariant along every guarded path. Every target state has a marked pair $\{z, z + \Delta\}$ for some z , so its marked tokens are in one lane and physically adjacent. This proves necessity. Lemma 4.2 gives a terminating guarded path for every lane-adjacent state, proving sufficiency.

Corollary 5.2 (cyclic-branch classification; N3a).

For every integer s ,

$$\boxed{\text{SafeHit}_{n, \Delta, \psi_\Delta^s}(x) \iff \text{LaneAdj}_\Delta(x) = 1}. \quad (4.7)$$

Proof. Theorem 3.1 removes the exponent s from the Safe-Hit truth predicate, and Theorem 5.1 classifies the resulting base system.

This is a path-free target-reachability theorem. It does not identify a common witness word for different exponents, compare shortest paths, or make LaneAdj_Δ a deterministic quotient of the labelled partial system.

6 Normalizer Extension: Skew Quotient and Affine Form

Let

$$\mathcal{Z}_\Delta = \widehat{\Sigma}/H_\Delta, \quad \pi_H : \widehat{\Sigma} \rightarrow \mathcal{Z}_\Delta. \quad (5.1)$$

Suppose a normalizes H_Δ . It induces a deterministic orbit permutation

$$\bar{a}([x]) = [a_\# x]. \quad (5.2)$$

For the identity branch define the relation-valued edge

$$O R_r^{(0)} O' \iff \exists y \in O \cap \text{Dom}(F_r^{(0)}), \quad F_r^{(0)}(y) \in O'. \quad (5.3)$$

Theorem 6.1 (normalizer skew-orbit reduction; N4.1).

If $a \in N_{\text{Sym}(E)}(H_\Delta)$, then

$$\boxed{O \xrightarrow{r}_a O' \iff \bar{a}(O) R_r^{(0)} O',} \quad (5.4)$$

$$\boxed{\pi_H(\widehat{\mathcal{T}}_{\Delta,a}) = \bar{a}^{-1} \pi_H(\widehat{\mathcal{T}}_{\Delta,\text{id}}),} \quad (5.5)$$

and the quotient is exact for target reachability:

$$\boxed{\text{SafeHit}_{n,\Delta,a}(x) \iff \pi_H(x) \rightsquigarrow_a \pi_H(\widehat{\mathcal{T}}_{\Delta,a}).} \quad (5.6)$$

Here $\rightsquigarrow_a$ is generated by the relations (5.4), not by deterministic quotient maps.

Proof. For an edge witness $x \in O$, put $y = a_\# x$. The normalized return factors as

$$F_r^{(a)}(x) = F_r^{(0)}(y),$$

with exactly the identity-branch guard at y . Since $a_\# : O \rightarrow \bar{a}(O)$ is a bijection, this proves (5.4). The identity $\widehat{\mathcal{T}}_{\Delta,a} = a_\#^{-1} \widehat{\mathcal{T}}_{\Delta,\text{id}}$ gives (5.5).

Finally, $F_0^{(a)} = a_\#$ and $F_\Delta^{(a)} = (\psi_\Delta a)_\#$ are globally enabled permutations. Their positive powers realize inverses, and their relative product realizes ψ_Δ . Hence every H_Δ -orbit is internally strongly connected by actual return blocks. The usual relation-valued orbit-lifting argument proves (5.6).

The twist $\bar{a}$ occurs before every base relation in (5.4). The theorem does not identify the normalizer system with the identity-branch system.

Write

$$M = \text{ord}(\psi_\Delta) = \begin{cases} \ell - 1, & g = 1, \\ \ell(\ell - 1), & g > 1. \end{cases} \quad (5.7)$$

Theorem 6.2 (affine-lane normal form; N4.2).

Choose a coordinate $t \in \mathbb{Z}/(\ell-1)\mathbb{Z}$ on the punctured lane C_* and coordinates (j, t) , $t \in \mathbb{Z}/\ell\mathbb{Z}$, on the ordinary lanes C_j , $j \in J = \{1, \dots, g-1\}$. A permutation a normalizes H_Δ if and only if there are

$$u \in (\mathbb{Z}/M\mathbb{Z})^\times, \quad \sigma \in \text{Sym}(J), \quad \beta_* \in \mathbb{Z}/(\ell-1)\mathbb{Z}, \quad \beta_j \in \mathbb{Z}/\ell\mathbb{Z} \quad (5.8)$$

such that

$$\boxed{\begin{aligned} a(*, t) &= (*, ut + \beta_*) & (\text{mod } \ell - 1), \\ a(j, t) &= (\sigma(j), ut + \beta_j) & (\text{mod } \ell). \end{aligned}} \quad (5.9)$$

When $g = 1$, the second line and σ are absent.

Proof. A normalizer element satisfies $a\psi_\Delta a^{-1} = \psi_\Delta^u$ for a unique unit $u \pmod{M}$. The power ψ_Δ^u has the same cycle decomposition as ψ_Δ . Therefore a fixes the unique punctured lane and permutes the ordinary lanes. In coordinates where ψ_Δ adds one, the identity $a\psi_\Delta = \psi_\Delta^u a$ forces (5.9), with one free phase on each lane. Conversely, every map of the displayed form is bijective and satisfies that identity.

7 Orientation-Compatible Normalizers and the Sandwich

The following simulation does not require normalizer membership.

Lemma 7.1 (identity-branch simulation; N4.3).

For every $a \in \text{Sym}(E)$,

$$\boxed{\text{SafeHit}_{n,\Delta,\text{id}}(x) \implies \text{SafeHit}_{n,\Delta,a}(x).} \quad (6.1)$$

Proof. Let $m = \text{ord}(a)$. The label-zero return is the globally enabled permutation $F_0^{(a)} = a_\#$. Before simulating an enabled identity-branch edge with label r , apply 0^{m-1} to reach $(a^{-1})_\# x$ and then apply r . The result is exactly $F_r^{(0)}(x)$. Repeat this edge by edge. If the identity path ends at $y \in \widehat{\mathcal{T}}_{\Delta,\text{id}}$, one final 0^{m-1} block reaches $(a^{-1})_\# y \in \widehat{\mathcal{T}}_{\Delta,a}$.

Let $\text{SameLane}_\Delta(x)$ mean that the two marked points lie in one ψ_Δ -lane.

Corollary 7.2 (same-lane obstruction; N4.4).

For every normalizer branch,

$$\boxed{\neg \text{SameLane}_\Delta(x) \implies \neg \text{SafeHit}_{n,\Delta,a}(x).} \quad (6.2)$$

Proof. A normalizer permutes whole lanes, and every guarded base return translates all lane indices uniformly by Lemma 4.1. Hence every normalized a -return preserves whether the two marked points share a lane. Every point of $\widehat{\mathcal{T}}_{\Delta,a} = a_\#^{-1} \widehat{\mathcal{T}}_{\Delta,\text{id}}$ has the same-lane property.

Theorem 7.3 (orientation-compatible normalizer classification; N4.5).

Let a have multiplier u in Theorem 6.2. If

$$u \equiv 1 \pmod{M} \quad \text{or} \quad u \equiv -1 \pmod{M}, \quad (6.3)$$

then

$$\boxed{\text{SafeHit}_{n,\Delta,a}(x) \iff \text{LaneAdj}_\Delta(x) = 1.} \quad (6.4)$$

Proof. Formula (5.9) shows that a preserves or reverses the cyclic order on every lane, while possibly permuting lanes and changing phases. Thus a preserves lane adjacency. Lemma 4.1 implies that every guarded a -return preserves the same observable. The target $a_\#^{-1} \widehat{\mathcal{T}}_{\Delta,\text{id}}$ is lane-adjacent, so adjacency is necessary. If the start is lane-adjacent, Theorem 5.1 gives an identity-branch Safe-Hit path and Lemma 7.1 simulates it in the a -system.

The full normalizer is nevertheless trapped between two intrinsic observables.

Corollary 7.4 (normalizer Safe-Hit sandwich; N4.6).

For every $a \in N_{\text{Sym}(E)}(H_\Delta)$,

$$\boxed{\text{LaneAdj}_\Delta(x) = 1 \implies \text{SafeHit}_{n,\Delta,a}(x) \implies \text{SameLane}_\Delta(x)}. \quad (6.5)$$

Proof. The first implication is Theorem 5.1 followed by Lemma 7.1. The second is Corollary 7.2.

Thus any new Safe-Hit state created by a multiplier $u \not\equiv \pm 1 \pmod{M}$ must already have its marked points in one lane but not adjacent in the hole-deleted token order. Corollary 7.4 does not assert that any such state is hittable.

8 Boundaries and Open Skew Multipliers

The normalized object has separated three levels that should not be merged.

1. The full parameter $a \in \text{Sym}(E)$ determines a guarded partial system, not merely a return group.
2. The cyclic branch admits the complete criterion of Corollary 5.2.
3. Normalizer membership gives the skew quotient of Theorem 6.1, but only the multipliers $u \equiv \pm 1$ are classified here.

For $u \not\equiv \pm 1$, multiplication inside a lane can change the hole-deleted five-token order. The remaining problem is exactly the skew dynamics on

$$(\mathcal{Z}_\Delta, \bar{a}, \{R_r^{(0)}\}), \quad (7.1)$$

restricted by Corollary 7.4 to same-lane, nonadjacent starts. The affine data (u, σ, β) may affect this reachability. No claim is made that the multiplier alone determines Safe-Hit truth.

The broader regime in which a merely preserves the ψ_Δ -cycle partition is also open. There the lane-wise permutations need not have a common multiplier, and even the normalizer skew form need not survive.

After pair hitting comes a separate survivor-incidence problem. The present theorems decide whether the marked source pair can reach the terminal kernel while all three spectators remain collision free. They do not determine the three survivor coordinates after the strict defect event.

Finally, raw reachability does not provide the typed observables needed for transfer membership or projectability. Any raw-to-typed audit must use the existing interfaces and stop at the first missing all- n clause; this paper introduces no replacement semantics.

9 Computational Artifacts

The paper-owned evidence package is available under [experiments/paper31/](#) in the RIME repository. It has no dependency on the broader exploratory source tree.

Surface	Paper-owned path	Role
Bounded hostile controls	<code>partial_return_audit.py</code> and <code>results/</code>	Replays the retained cyclic and normalizer domains
Partial formalization	<code>lean/</code>	Checks the generic path, quotient, skew-transport, and boundary spine
Release validation	<code>release-manifest.json</code> and the public receipt under <code>results/</code>	Binds the manuscript, reader PDF, bibliography, nested development closure, environment, and validators

The retained audit covers 93,720 cyclic-branch states, 870,540 enabled labelled edges, 456 normalizer branches, 70,560 normalizer starts, 36,760 orientation-compatible cases, and 175,216 quotient edges. The Lean spine checks only the declarations listed in its README; it does not formalize the punctured-lane arithmetic or constructive hole-slide. These are bounded consistency and formalization controls, not proofs of the all- n theorems or independent Computational Certificates.

The release manifest binds the canonical reader package and its nested development closure. The public receipt records performed replay and local closure verification, is excluded from its own closure, and is not an independent mathematical validation.

10 Claim Status and Boundary

Claim surface	Status	Scope
Branch-normalized partial-system isomorphism	Theorem 2.1	Formula, exact guard and target, and label translation for every oriented binary-kernel rank- $(n - 1)$ defect, $n \geq 6$
Cyclic-branch exponent elimination	Theorem 3.1	Every $a = \psi_{\Delta}^s$ and every start state
Lane-order preservation	Lemma 4.1	Every guarded identity-branch return
Terminating lane hole-slide	Lemma 4.2	Every lane-adjacent marked pair
Identity-branch Safe-Hit criterion	Theorem 5.1	All $n \geq 6$ and all kernel separations Δ
Complete cyclic-branch classification	Corollary 5.2	Every $a \in \langle \psi_{\Delta} \rangle$
Exact normalizer skew quotient	Theorem 6.1	Relation-valued orbit reduction, not deterministic descent
Affine lane normal form	Theorem 6.2	Every normalizer branch
Identity-branch simulation	Lemma 7.1	Every branch permutation a
Same-lane obstruction	Corollary 7.2	Every normalizer branch
Orientation-compatible classification	Theorem 7.3	Normalizer multipliers $u \equiv \pm 1 \pmod{M}$
Normalizer Safe-Hit sandwich	Corollary 7.4	Every normalizer branch
Other normalizer multipliers	Open	Same-lane, nonadjacent frontier only
Cycle-partition-preserving regime	Open	No general Safe-Hit criterion claimed
Survivor placement	Deferred	Terminal off-kernel incidence not classified here
Raw-to-typed bridge	Open	No new semantic fields introduced
Projectable origins and recursive return	Deferred	Requires later typed and same-witness theorems

All positive claims above have direct mathematical proofs in the manuscript. The bounded finite audit and Lean spine are consistency and formalization controls; neither is a theorem premise or an independent validation of those proofs.

The following implications are explicitly excluded:

$$a \in N_{\text{Sym}(E)}(H_{\Delta}) \implies \text{SafeHit} = \text{LaneAdj}, \quad (7.2)$$

$$\text{SameLane}_{\Delta}(x) = 1 \implies \text{SafeHit}_{n,\Delta,a}(x), \quad (7.3)$$

and

$$\text{raw Safe-Hit} \implies \text{typed transfer membership or projectability}. \quad (7.4)$$

11 Conclusion

The full return dynamics of an arbitrary labelled single-defect circular automaton admits an exact branch normalization. The formula $\phi_r = \varepsilon_\Delta p^r a$ carries the collision guard, section exit target, and labelled paths together, reducing the intrinsic raw problem to the parameters (n, Δ, a) .

For the cyclic branch, the exponent disappears after relation-valued orbit reduction. The remaining base dynamics has a complete all- n classification: the marked pair reaches the exit target exactly when its tokens occupy one punctured-rotation lane and are adjacent after holes are deleted. A guarded hole-slide supplies a terminating witness path.

Beyond the cyclic branch, normalizer dynamics is an exact skew system over the identity-branch relations. Its affine lane multiplier identifies the orientation-compatible subregime $u \equiv \pm 1$, where the same classification survives. For the full normalizer, the Safe-Hit sandwich confines every open case to a same-lane, nonadjacent start. The next problem is therefore no longer an unspecified general defect: it is the arithmetic skew dynamics of the remaining affine multipliers.

References

- [1] Mikhail V. Berlinkov and Cyril Nicaud. Synchronizing almost-group automata. *International Journal of Foundations of Computer Science*, 31(8):1091–1112, 2020.
- [2] David Fernando Casas Torres. Completely reachable almost group automata, 2024.
- [3] Ján Černý. Poznámka k homogénnym experimentom s konečnými automatmi. *Matematicko-fyzikálny časopis Slovenskej Akadémie Vied*, 14:208–216, 1964.
- [4] WuJun Chen. Cyclic lineage dynamics and rank-five first-exit frontiers. RIME Paper XXIX, version 1.0, Zenodo, September 2026. DOI: <https://doi.org/10.5281/zenodo.23028361>.
- [5] WuJun Chen. Universal return groups and punctured rotations. RIME Paper XXX, version 1.0, Zenodo, September 2026. DOI: <https://doi.org/10.5281/zenodo.23034428>.
- [6] Henk Don. The Černý conjecture and 1-contracting automata. *The Electronic Journal of Combinatorics*, 23(3):P3.12, 2016.
- [7] Lucien Dubuc. Sur les automates circulaires et la conjecture de Černý. *RAIRO Informatique théorique et applications*, 32:21–34, 1998.
- [8] David Eppstein. Reset sequences for monotonic automata. *SIAM Journal on Computing*, 19(3):500–510, 1990.
- [9] Robert Ferens and Marek Szykuła. Recognizing completely reachable automata in quadratic time. *ACM Transactions on Algorithms*, 22(2):1–36, 2026.
- [10] François Gonze and Raphaël M. Jungers. On completely reachable automata and subset reachability. In *Developments in Language Theory*, volume 11088 of *Lecture Notes in Computer Science*, pages 330–341. Springer, 2018.
- [11] Andrew Ryzhikov and Anton Shemyakov. Subset synchronization in monotonic automata. *Fundamenta Informaticae*, 162(2–3):205–221, 2018.
- [12] Mikhail V. Volkov. Synchronizing automata and the Černý conjecture. In *Language and Automata Theory and Applications*, volume 5196 of *Lecture Notes in Computer Science*, pages 11–27. Springer, 2008.

- [13] Mikhail V. Volkov. Synchronization of finite automata. *Russian Mathematical Surveys*, 77(5):819–891, 2022.
- [14] Petra Wolf. Synchronization under dynamic constraints. In *40th IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science (FSTTCS 2020)*, volume 182 of *Leibniz International Proceedings in Informatics*, pages 58:1–58:14. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2020.